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1. Title of the invention:

Aegis - Multilayer Smart Security System Using Sensor Fusion and ML Classification

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Embedded Systems Architecture (BACSE324) | Slot E1+TE1

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2. Field /Area of invention:

Embedded presence monitoring using a combination of PIR and ultrasonic sensing, triggered camera capture, person detection via a classification model, event-aware alarm feedback and clearance-based controller rearming.

3. Prior Patents and Publications from literature

A. Prior patent literature

Reference / citation	What the work discloses	Relevance to Aegis	Technical comparison
P1. US5473368A, Interactive surveillance device, 1995	PIR sensing, a camera and ultrasonic ranging are combined in an interactive surveillance security device monitoring a premise	Shows earlier integration of the same broad and large sensing modalities shown to detect intruder presence	Aegis uses a fixed perimeter, three consecutive motion and distance confirmation and software ML based person detection. Merely combining these components is not claimed as a new invention
P2. US9922514B2, Apparatus and methods for alarm verification based on image analytics, 2018	Image analysis assist the verification of sensor generated alarms	Establishes visual verification as known prior work	Aegis documents a specific local ESP32 capture via ESP32 CAM queued inference and two duration of buzzer workflow. No conclusion about claim coverage is made
P3. US20170076588A1, Intrusion detection with motion sensing, 2017	Discusses combined video-motion and PIR detection, including AND logic to reduce false alarms	Shows that logical agreement between different modalities is an established approach	Aegis uses PIR motion and measured distance for its first layer, followed by image detection. Consecutive counting and rearming are implementation choices, not proof of patent novelty

B. Research publications

Reference / citation	What the work discloses	Relevance to Aegis	Technical comparison
R1. Pugh, Brewer and Kvam, 2015. doi:10.1109/SAS.2015.7133634	Multiple-PIR room intrusion detection with a false-alarm constraint	This provides motivation for evaluating detection and false alarm performance in combination	Aegis includes ultrasonic distance sensing and camera person detection. Different datasets and tasks were used
R2. Ciuffreda, Casaccia and Revel, 2023. doi:10.3390/s23156963	Indoor Robot Person Localization Using PIR and Ultrasound Sensors	This provides additional physical sensors to complement each other.	Aegis monitors a single area. It combines captures, visual analysis and acoustic feedback.
R3. Patino et al., 2022. doi:10.3390/s22197351	Heterogeneous sensor fusion in border surveillance	This paper supports having multiple stages of interpretation with heterogeneous sensors.	Aegis is a small localised version. No direct comparative evaluation was conducted to establish Aegis's performance relevant to this system's work

Prior-art boundary: combined sensing, AND logic, triggered imaging and visual alarm verification are already known. The report describes an engineering integration with specific confirmation, event-handling and reset rules. The selected references do not establish a novel patentable combination, and they do not provide a common benchmark for the prototype.

4. Summary and Background of the Invention (Addressing the gap / Novelty)

Background and gap

Motion detection alone does not confirm that a visible person is there. Distance detection contributes spatial information, while camera processing brings visual information into the mix of the system. Any prototype will also have to account for false readings, missing uploads, prediction delays, and repeated alarm responses instead of relying on the presumption that one trigger signal proves full detection.

Aegis uses PIR motion detection and an ultrasonic reading within an acceptable range for at least three consecutive intervals. The controller sends a request for capturing a picture and sounding a short beep. The image is processed with a person detector by a ML model which is processed in the computer and stored, with the result made available for the controller's polling . A clearance mechanism prevents repeated captures until the monitored region becomes empty.

Contribution and scope

The contributions of Aegis includes the integration of dual sensing, original image storage, queuing and local reasoning within the main device, duplicate removal from feedback, two auditory alarm response phases with buzzer, and independent rearming. The proposed advantage is fewer false alarms while tolerating missed events. This advantage has not been quantitatively compared against PIR only, dual sensor and complete system scenarios.

5. Objectives

To utilise both motion and distance sensors in conjunction with each other so as to reduce false positives.

To confirm a sensor event using three consecutive readings that satisfies the condition of distance and motion sensor

Retain original images from the camera. Keep records for person detection boxes, confidence levels and prediction values.

No indication to ensure that the system has successfully processed end to end. 1 second beep from sensor fusion to indicate an unknown person has passed by. 5 second alert for ML classification's successful detection

To reject stale or duplicate responses and rearm the system after stable clearance of the monitored area.

Missed detections, false alerts, upload consistency, upload latency can be tested using identical physical testing.

6. Principle of Operation

While armed, the system checks for valid PIR motion or a valid ultrasonic distance of 80cm or less.

For every consistent reading increment the counter. If inconsistent or an invalid echo reset the counter.

If 3 consistent readings have been received, it triggers a 500ms camera pulse and 1 second sensor beep. The system will now wait to be cleared.

ESP32-CAM will take up to 5 JPEG images and send them to the laptop backend to be processed.

The backend stores original images. SQLite tasks are queued. The YOLO classification model receives and processes each image.

Person status, confidence level, person detection boxes, number of detected people, inference time and annotated image are stored for the dashboard.

Main ESP32 will wait for a new person's positive event. If a new valid event occurs, it sends out a 5 second alert.

After the trigger pulse, the system will rearm once the PIR is clear and a valid distance (>100cm) has been maintained for 2 seconds. If any of the conditions fail to be clear the timer will reset to stay in WAIT_CLEAR state.

The laptop will perform ML inference. Neither ESP32 boards will run YOLO. The model will detect a visible 'person' class but cannot identify who the person is, authorise them or determine their intent or liveness. The sensor will rearm and ML will process independently in the main device (ie. the laptop).

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7) Detailed description of invention (Drawings and/or images below if necessary)

7.1 Architecture & workflow

The first layer consists of the PIR and ultrasonic sensors connecting to the main ESP32. The wired output then activates a separate ESP32-CAM unit. The second layer is the upload of the camera output of burst Images to the laptop running the receiver, resilient job queue, ML worker and Aegis dashboard. The controller then queries the laptop for feedback on alarm status and controls a singular passive buzzer.

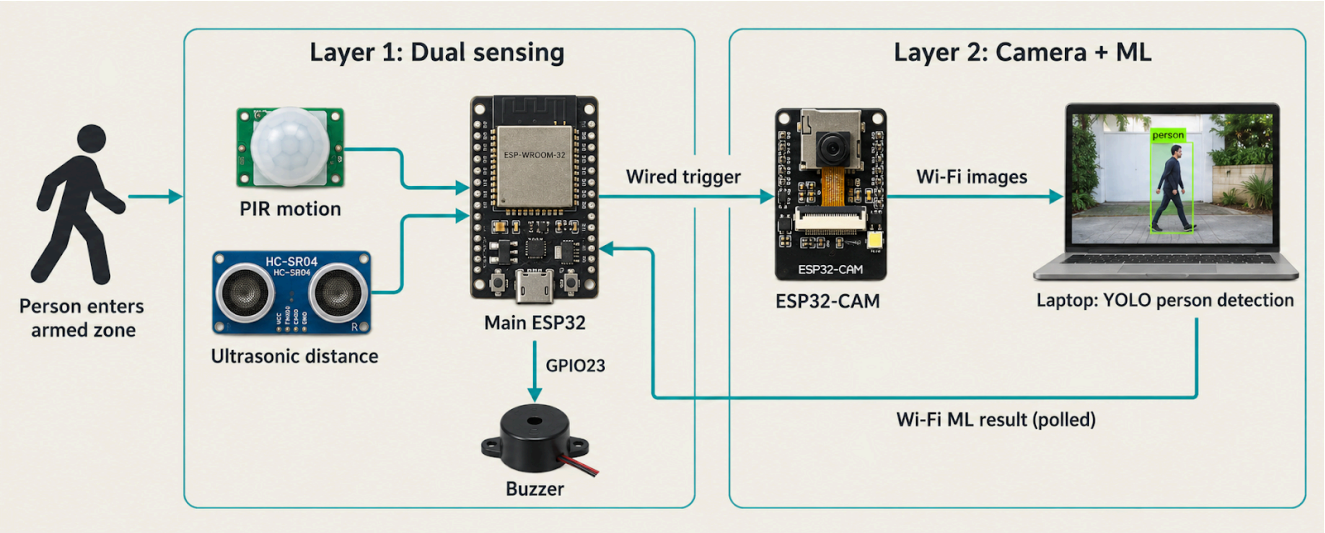


Figure 1. Aegis architecture and action sequence performed. Conceptual diagram shown above

Component or connection	Implementation
PIR output	Main GPIO13
Ultrasonic trigger	Main GPIO5 echo through a voltage divider
Ultrasonic echo	GPIO18 , echo through a voltage divider
Camera trigger	Main GPIO19 to camera GPIO13
Passive buzzer	Main GPIO23 to buzzer; shared ground
Camera and laptop	Local Wi-Fi image upload for main device(ie laptop) ML worker
Supply	Regulated supply and common ground; camera supply stability requires verification

7.2 Dual-sensor fusion table

Distance measurement

The ultrasonic sensor calculates the distance using the sound pulse's round trip travel time (RTT). It uses the formula d = c × t / 2, where d is distance, c is the speed of sound (343 m/s), and t is the measured echo travel time. The distance is divided by 2 since the signal travels to and from the object.

$$d\ (cm) \approx 0.0343 \times t\ (\mu s) / 2$$

This calculation is done only for valid echoes. An invalid echo for the ultrasonic sensor means unknown distance, not a clear area.

PIR	Ultrasonic	Agreement count	Automatic action while armed
No motion	Valid >80 cm	Reset to 0	Continue monitoring
Motion	Valid >80 cm	Reset to 0	No sensor-triggered capture
No motion	Valid <=80 cm	Reset to 0	Nearby object alone does not trigger
Motion	Valid <=80 cm	Increment	At 3 consecutive confirmation then capture trigger and 1-second beep
Either	No echo or invalid	Reset to 0	No new sensor event

Sensor agreement and trigger condition

For a given sample condition, when the PIR detects motion as well as the ultrasonic sensor detects an object within 80cm, it triggers an agreement. This uses a simple AND logic. For each agreement, the agreement counter increments.

While ARMED, reaching counter = 3 triggers the ESP32 camera pulse and requests a one second buzzer beep. The 3 triggers refer to 3 consecutive sensor matches, but do not reflect on the time delay for the triggers.

The ultrasonic sensor has a validation stage. If the echo is invalid then the distance will be unknown to the device. Having three echoes means that the sampling occurs on consecutive occasions for the area. This means that if there are three matches, then the ultrasound echoes are validated. If the echo is invalid, or the distance is above the threshold then the confirmation counter is reset.

7.3 Camera Capture & Storage

The camera is implemented using OV3660 with resolution 800x600, PSRAM, quality of 12 JPEG. It will only have one frame buffer and take a pause between images of 250ms. Full trigger will acquire up to 5 frames. Images will be captured one at a time waiting for an ack. Any retries will have the same bytes and id. The upload handler will validate the token, id, JPEG and decoding. The original image will be stored.

7.4 Human Detection & Filtering

The worker will be using yolo11m.pt with threshold 0.35 and 960 inference size. OpenCV will decode each image. CLAHE has been turned on to increase local contrast in darker images. The image will be resized but will not have any real detail. Any boxes must be above 0.35 confidence, 0.5% of the area of the image, 10% of the height of the image and have a width/height ratio of less than 3.0.

If there are no valid boxes then padding may be turned on (if enabled). Duplicate boxes will be filtered out. TTA is disabled by default. These conditions may cause some false positives to be rejected but could also cause real small or partial humans to be rejected. If no boxes are valid then a 0 will be returned.

7.5 Processing & Tracing

Only one worker will load and warm up the model. SQLite will log the queue, processing and final status. The worker will write the annotated image (JPEG) and JSON. It will use an exclusive lock and publish a heartbeat. Any jobs that didn't finish processing can be queued again when starting. If there is an error during inference then it will still be considered an error. Aegis will group the frames into bursts and display the original and annotated images.

7.6 Alerts

If an event is confirmed by the sensor then it will produce a 1second alert regardless of Wi-Fi connection. The main controller will check with the backend every second. Events must have been received and inference completed in the last 90 seconds to be considered a new and positive person event. The first successful boot response will cause a silent alert to occur. The final processed event ID ensures duplicate responses will not occur.

If a new event occurs which is an ML event then a 5s alert will occur. This will override any short alerts of the buzzer. Any further sensor events will not cause the alert to be shortened or reset whilst an ML alert is occurring. No ML alerts will be produced for no-person events, pending events, failed events, stale events or legacy imports. The alert expiry is controlled by the timer task. The alarm duration is separate from the delay between the sensor being triggered and the alarm being activated.

7.7 Re-arm and measurement

The WAIT_CLEAR state will change to ARMED once the 500ms pulse has finished, if the PIR readings show clear indicating no motion and the distance remains valid and above 100cm for 2 seconds. If there is movement, distance is less than or equal to 100cm or an invalid echo the clear timer will reset. No waiting for inference or buzzer to finish. The trigger distance is 80cm and clear distance is 100cm which helps reduce double triggering.

Rearming condition and hysteresis

After the 500 ms trigger pulse has ended, the clearance condition is as stated:

$R(t) = [PIR(t) = 0]$ AND $[V(t) = 1]$ AND $[d(t) > 100\ cm]$

PIR = 0 means no motion,

V = 1 means a valid ultrasonic echo,

and d is measured distance.

Return to ARMED only when R(t) remains TRUE continuously and repeatedly for more than 2 seconds. Any failed condition resets the clearance timer.

Hysteresis gap = clearance threshold − trigger threshold = 100 − 80 = 20 cm

This 20 cm gap separates the trigger and clearance boundaries, helping prevent repeated triggering near the boundary. Rearming is independent of ML processing & classification and buzzer alert completion time.

7.8 Finite State Machine (FSM)

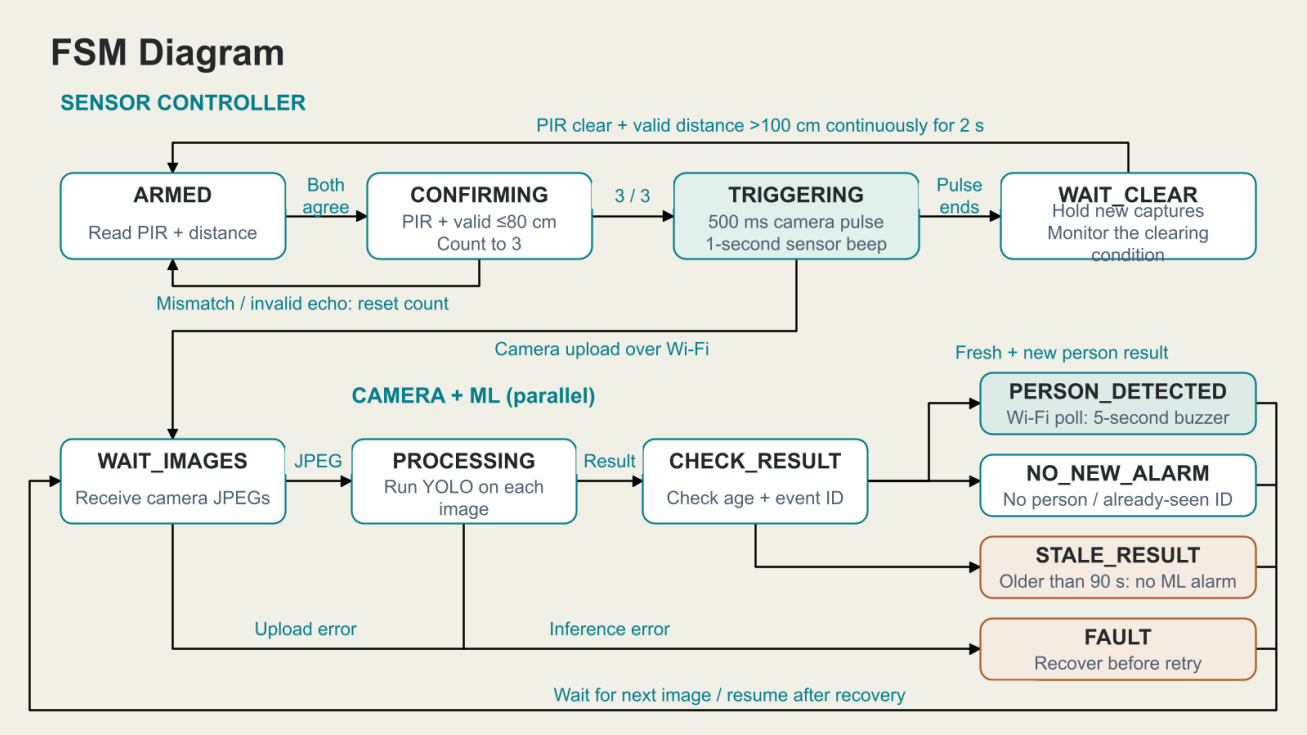


Figure 1a. Controller states for sensor triggering and rearming, with independent ML alarm feedback.

This conceptual FSM shows the sensor controller and camera & ML workflow running in parallel. Sensor rearming depends on stable clearance within the premises of the system and does not wait for ML processing or buzzer completion. Fresh people detected by the ML model can request a 5 second alarm, repeated no person and stale results add no ML alarm. Faults require recovery before processing resumes.

7.9 Camera specific improvements

A separate labelling tool has been set up to manually label person, empty scenes, uncertain, time and original hash. There were around a thousand JPEG images captured and 325 annotations saved. Of these 166 were labelled as persons and 159 as empty.

The exporter has been prepared to only include labelled persons and empty images. Uncertain and unlabelled items will be excluded. Hashes will be checked. Duplicates will be grouped together. Each session will be grouped together. The aim is to have 70% training images, 15% validation and 15% testing images. All splits will contain positive and empty images. Grouping by session cannot confirm that scenes are independent.

The candidate run has been prepared. Starting from the YOLO11m model, 50 epochs maximum, patience 10, training size 640, batch size 2 and seed 42. Validation will be carried out using threshold values of 0.25, 0.35, 0.45 and 0.55. Using the production detector with size 960. The dataset has been partially exported, trained or compared. The labelled setup is CPU only.

7.10 Limitations

Low lighting, motion blur, seated or partly visible people, unusual camera angles, blocked lenses, sensor coverage and strict detection filters can cause missed detections. A sensor beep alone does not confirm that an image was captured, uploaded or processed.

The person-detection model identifies visible people but does not determine whether they are authorized to enter. Distinguishing authorized personnel from unauthorized visitors requires a separate authorization check, such as a badge or QR code checked against an approved personnel registry.

8. Experimental validation results:

8.1 Saved image-level evaluations

Evidence	Configuration	Observed counts	Interpretation
Historical 227-image report	person_ft.pt, threshold 0.50	TP 137, FN 7, FP 2, TN 81	Precision 98.56%; recall 95.14%; FPR 2.41%. Historical machine/report.
Recovered 218-image report	person_ft.pt, threshold 0.50	TP 129, FN 7, FP 2, TN 80	Precision 98.47%; recall 94.85%; FPR 2.44%. Independence from training unproven.

Performance calculations — historical 227-image evaluation

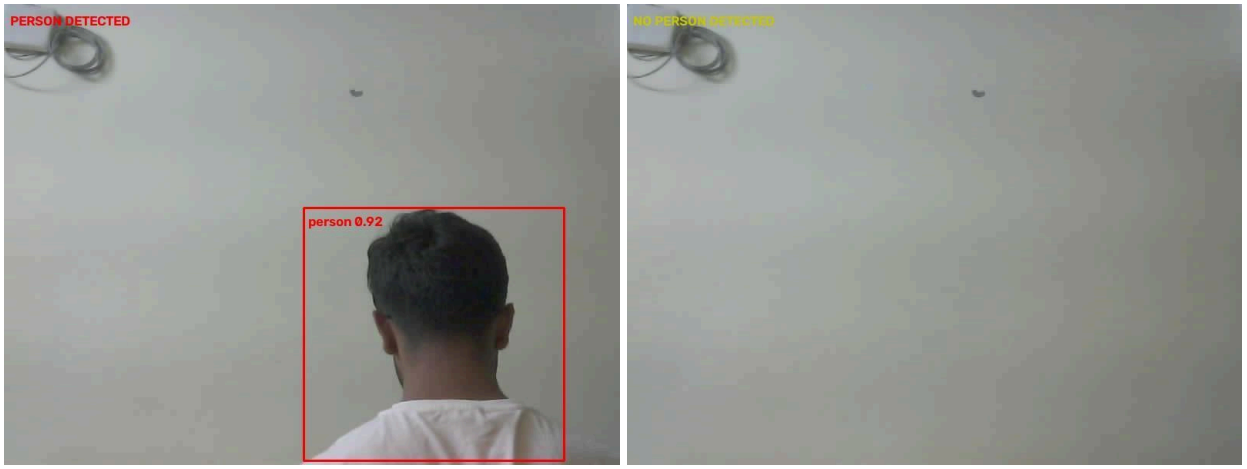
For this historical person_ft.pt evaluation at threshold 0.50: TP = 137, FN = 7, FP = 2 and TN = 81. The total N = 227. TP means a person image correctly detected out, FN is a person image missed,FP is a no-person image incorrectly flagged, TN is a no-person image correctly rejected all out of the sample image dataset provided.

Metric	Formula (%)	Calculation	Result
Precision	TP / (TP + FP) × 100	137 / 139 × 100	**98.56%**
Recall	TP / (TP + FN) × 100	137 / 144 × 100	**95.14%**
False-positive rate	FP / (FP + TN) × 100	2 / 83 × 100	**2.41%**
Accuracy	(TP + TN) / N × 100	218 / 227 × 100	**96.04%**

These calculations apply only to the historical 227 image based evaluation, not to the current YOLO11m deployment or the complete hardware system of the project. Person positive and no-person positives images form different denominators for recall and false-positive rate.

Checkpoint is slightly different to the current pretrained camera model. These figures cannot be assumed to reflect those of the deployed YOLO11m as slight fine tuning is needed in the process. Average inference time from initial camera image burst results was 523ms. No capture, transfer, queuing or polling delay is also included.

8.2 Saved camera examples



Figures 2 and 3. Later saved person frame at 91.90% detection confidence and a separate no-person frame with no accepted box.

With the perfect match, the person example has a confidence level of 0.92 and one match. The blank example with no person present had zero matches.The reported score represents the ML model's confidence for that prediction, it does not determine the overall accuracy of the system. Saved detection examples illustrate the model output, while the photographs in section 8.3 document the physical prototype of the overall setup.

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8.3 Physical prototype photographs

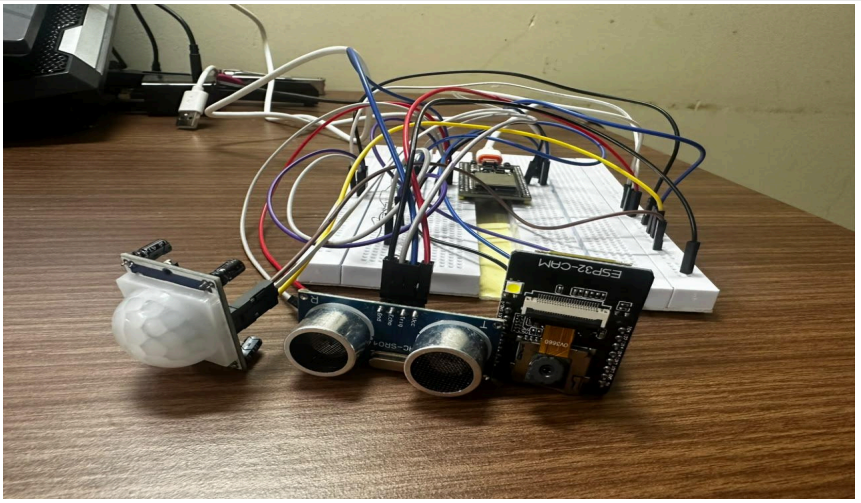


Figure 4. Aegis prototype overview showing the PIR sensor, ultrasonic sensor, ESP32-CAM and main ESP32 on the breadboard.

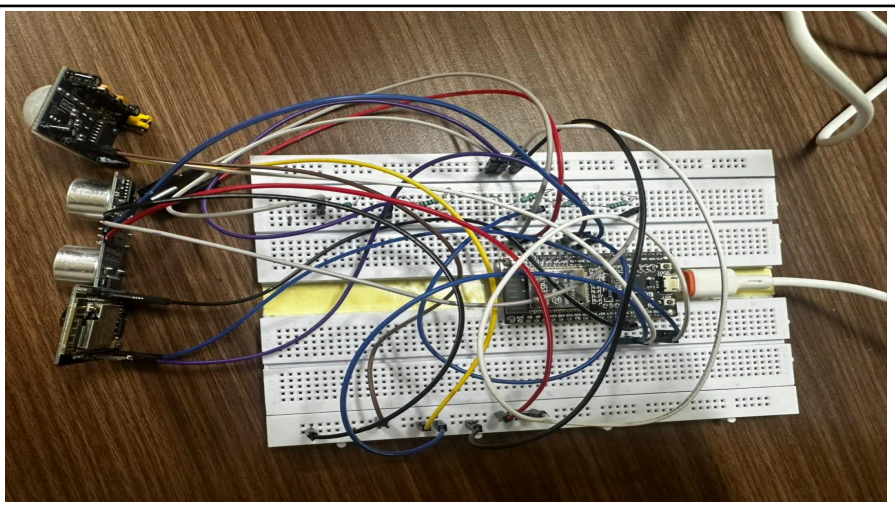


Figure 5. Top-down view of the main ESP32, breadboard wiring and connected sensing and camera modules.

8.4 Physical trial record

Observed parameter	Trial 1	Trial 2
Date / scene / event ID	Animal Motion	Unknown Person Presence
Sensor agreement and short beep	Sensor Agrees and beeps	Sensor Agrees and beeps
Images received / requested	4-5 Images	4-5 Images
ML verdict and inference time	No Person Detected in maximum 6000 ms.	Person Detected in maximum 5000 ms .
ML alarm request / audible result	No alarm request	Alarm requested via Wi-Fi for a span of 5 seconds.
Clearance and return to ARMED	The system rearms after the animal clears within the premise of the system.	System rearms after the person is cleared off the premise, which is indicated from the sensor clearance criteria.
Failures or interruptions	Stagnant animal movements	Distorted or unclear images.

9. What part(s) of your invention should be protected?

Whilst there are several relationships identified below to consider from an engineering perspective, this report does not claim patentability or protection of general sensors, logic, YOLO or visual verification for alarms. The relationships are:

1. Synchronisation between sequential PIR motion/dist sensor, capturing and initial audio (buzzer) alarm.

2. Separation of original camera footage, priority queued visual inferences and grouped event recordings for the photos.

3. Checking for new and old events to link asynchronous predictions on the laptop to a second alarm.

4. Priority given to short sensor and long ML buzzer windows.

5. Separate rearming based on clearance, rejecting false echoes.

Overall, this invention has a relationship between sensing, verifiable imagery, event based feedback and reliable physical clearance.The prototype has been validated in a laboratory supporting TRL4, Further testing in realistic operating environments is needed to assess the higher readiness and stability of the system. Additional prior-art review is needed before making strong novelty claims.

10. What is the technology readiness level of your invention?

Research			**Development**			**Deployment**		
TRL 1	TRL 2	TRL 3	TRL 4	TRL 5	TRL 6	TRL 7	TRL 8	TRL 9
Basic principles observed	Concept formulated	Experimental proof of concept	Validated in a laboratory	Validated in a relevant environment	Demonstrated in a relevant environment	Prototype in an operational environment	Complete and qualified	Proven in operational use
			✓					

Proposed TRL: TRL 4 – Validated in a laboratory (subject to faculty confirmation)

The Aegis prototype has been validated in a laboratory, carefully explained and demo tests have been conducted. This supports a proposed classification of TRL 4. The physical trial record in Section 8.4 describes the observed system behaviour. This classification refers to laboratory validation and does not establish validation in an operational environment.

Reference sources and supporting records

P1. US5473368A. Interactive surveillance device. 1995. https://patents.google.com/patent/US5473368A/en

P2. US9922514B2. Apparatus and methods for alarm verification based on image analytics. 2018. https://patents.google.com/patent/US9922514B2/en

P3. US20170076588A1. Intrusion detection with motion sensing. 2017. https://patents.google.com/patent/US20170076588A1/en

R1. M. Pugh, J. Brewer and J. Kvam. Sensor Fusion for Intrusion Detection Under False Alarm Constraints. IEEE SAS, 2015. doi:10.1109/SAS.2015.7133634.

R2. I. Ciuffreda, S. Casaccia and G. M. Revel. A Multi-Sensor Fusion Approach Based on PIR and Ultrasonic Sensors Installed on a Robot to Localise People in Indoor Environments. Sensors, 23, 6963, 2023. doi:10.3390/s23156963.

R3. L. Patino et al. Fusion of Heterogenous Sensor Data in Border Surveillance. Sensors, 22, 7351, 2022. doi:10.3390/s22197351.

Project evidence: ml/results/report.txt; ml/results_current/report.json; ml/evaluation/esp32_camera_report.json; matching gallery prediction JSON; detector.py; backend/worker.py and storage.py; main_ml_alarm firmware; docs/ml-buzzer.md and camera-finetuning.md.

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